

Smart Aquaponics IoT System Design: Vertical Approach

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Abstract: A Smart Aquaponics IoT System with a vertical approach is an innovative and efficient method for sustainable food production, integrating fish farming with hydroponic plant cultivation while leveraging Internet of Things (IoT) technology for automation. This system utilizes vertically stacked grow beds or NFT (Nutrient Film Technique) towers to maximize space efficiency, making it ideal for urban and small-scale farming. IoT sensors continuously monitor essential parameters such as pH, temperature, dissolved oxygen, water level, and nutrient concentration, transmitting real-time data to a cloud-based platform for analysis. AI-driven automation optimizes water circulation, nutrient delivery, and aeration, ensuring an ideal environment for both fish and plants. This system design is intended to maximize space utilization, making it perfect for rooftops, greenhouses, and urban settings. It significantly reduces water consumption (nearly up to 90%) by continuously recycling water, making it more sustainable than traditional farming. Automated monitoring system by applying IoT technology is designed efficiency by reducing manual labour and allows to remotely monitor and control system conditions, improving convenience and precision. Additionally, the organic nutrient cycle eliminates the need for chemical fertilizers, ensuring eco-friendly and pesticide-free crop production. This smart, sustainable, and scalable solution is paving the way for high-yield urban agriculture.